DEEP REINFORCEMENT LEARNING FOR LOCALIZATION OF BRAIN LESIONS ON 2D-MRI

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ABSTRACT

Despite recent rapid success and performance of deep reinforcement learning (DRL) in multiple areas—including robotics, natural language processing, computer vision and gaming—there have been limited applications of DRL in medical imaging. It has been claimed that this may be due to the lack of systematic understanding of DRL's strengths and weaknesses when applied to medical data. Recently, there has been significant progress and advances in the use of Supervised Deep Learning (SDL) in radiology. More specifically, there have been many instances of radiologist-labeled data being used to train convolutional neural networks (CNNs); where, these CNNs have been levied to make predictions on unlabeled data for lesion classification, lesion segmentation and lesion localization. However, it can be argued that SDL approaches come with three major drawbacks for clinical translation in radiology: 1) the requirement of large-annotated data sets, 2) non-generalizable data due to differences in scanners and image acquisition protocol between sites, and 3) inadequate interpretability.

Reinforcement learning potentially can address many of these shortcomings with the use of robust intuitive algorithms trainable on small datasets. In this preliminary work, we aim to train a Deep Q-Network (DQN) to localize High Grade Gliomas (HGG) and Low Grade Gliomas (LGG) within 2D axial brain MRI slices, acquired from publicly available BraTS 2017-2020 datasets. This localization agent learns to define a bounding box using simple transformation actions with the ultimate goal of determining the most specific location for target brain tumors following top-down reasoning. Here we show that agents guided by the proposed model were able to localize instances of a brain tumors in T1-weighted contrast-enhanced (T_1-ce) 2D axial MRI images.

Our contributions in this paper are the following:

- We formulate the problem of tumor localization as a Markov decision process (MDP) since this setting provides a formal framework to model an agent that makes a sequence of decisions. We then apply a Deep Q-Network architecture on the task.
- We discuss the implementation of our localization agent and policy as well as the type of DRL Q-learning framework we used to implement.
- We experiment and analyze the performance of our proposed model in three different brain tumor categories or subtypes, HGG and LGG (T_1-ce) 2D axial MRI, LGG (T_1-ce) 2D axial MRI and HGG (T_1-ce) 2D axial MRI.

***K*eywords** Reinforcement Learning · MRI · Deep Q-Networks

1 Introduction

1.1 Background and Motivation:

Deep reinforcement learning (DRL) augments the reinforcement learning framework, which learns a sequence of actions that maximizes the expected reward, with the representative power of deep neural networks. Recently, there has been rapid progress in DRL resulting in dramatic performance improvement in many areas—namely games [1], natural language processing [2], robotics [3], and computer vision [4]. Unlike supervised learning (SL), the DRL framework can accommodate sequential decision making, and learn with delayed supervised information (i.e., the success or failure of the decision is available only after multiple time steps). DRL can also incorporate non-differentiable metrics in the form of network weights optimization, neuron activation function, and dropout techniques to name just a few which can maximize classification accuracy.

Despite the rapid growth of DRL implementations in many fields, DRL has not gained traction in the field of radiology compared to SL and other deep learning methods. Supervised Deep Learning (SDL) up until now has dominated AI implementations in radiology. In SDL, large numbers of explicitly labeled data are used to train convolutional neural networks (CNNs). These trained networks can then make predictions on new unlabeled data, often classification, semantic segmentation, or localization tasks [5, 6, 7, 8]. Recently, SDL has been complemented by unsupervised deep learning, which seeks to uncover commonalities and differences in data, without overt data labeling or annotation by a radiologist. Unsupervised deep learning is more of an exploratory procedure; typically, the researcher does not initially know what end-goal they are looking for specifically, and no goal can be provided to the algorithm [9, 10]. Currently SDL continues to constitute the vast majority of current radiology deep learning research.

However, it can be argued that SDL approaches within a radiological context suffer from major limitations. First, SDL requires significant amounts of expert annotated data—quite often a radiologist—which is laborious and time-consuming to acquire [11]. Second, SDL often is unstable in the sense of being sensitive to subtle differences in image quality. A small amount of MR detector or image noise can throw off the predictions of SDL [12]. As such, a network trained on a given patient population with certain MRI scanner settings at a particular institution in general fails to perform acceptably well when deployed on a new scanner and or different institution with different patient population. This makes it extremely difficult to perform large multi-site studies because often patient data is acquired on different MRI scanners from different manufacturers with completely different image acquisition parameters and requirements. Third, gaining the trust of people, particularly in a field as critical as health-care requires that the SDL algorithms help to make decisions in a transparent manner. Understanding the rationale for why SDL algorithms make certain determinations fulfills the basic needs of both patients and clinicians that is required before widescale adoption is possible [13].

DRL has the potential to address some of these drawbacks of these limitations. First, RL learns about the whole environment including much of the image that is not a region of interest or labeled by a radiologist [11]. This could potentially lead to fewer labeled images being needed for DRL implementations compared to SDL. Additionally, due to the discrete-action framework RL learning, where optimal policies are produced via the Markov Decision Process (MDP), RL approaches could be robust to noise and variations in image acquisition techniques and patient populations. This has to do with how RL works; goals are not provided explicitly as in SDL, but rather implicitly through a system of rewards. This process could potentially be a robust and a generalizable form of learning during training. [11, 14] However, this in practice could also not be the case. Finally, reward structures in RL provide some rationale for why an algorithm makes certain predictions. Despite this promise, RL has only been applied directly to medical images to our knowledge in a few instances [11, 14, 15].

Here, we aim to demonstrate a preliminary application of DRL in radiology. Our goal is to create an active detection model for localizing brain lesions in axial 2D-MRI images. The active detection lesion model allows an agent to focus attention on candidate regions for identifying the correct location of a brain tumor—in this case Low Grade Gilomas (LGG) or High Grade Gilomas (HGG). This agent learns to deform a bounding box using transformation actions, with the goal of determining the most specific location of the target. Potentially, this approach will lead to more sophisticated implementations for lesion detection, localization or even segmentation using RL.

1.2 Active Lesion Localization:

The process of localizing objects or targets in a 2D image with bounding boxes can be considered a control problem with a sequence of steps to redefine the geometry of the box. The accuracy of object detectors has been possible with the use of Convolutional Neural Networks (CNNs) that levy big visual data and deep learning to directly predict bounding boxes under regression setting also with good results [16, 17, 18].

In this work, we propose an active detection model that learns to localize target brain tumors known by the system, this model has been adapted from the general model proposed by Cicedo et al. [19]. The proposed model follows a top-down search strategy, meaning it begins by evaluating the whole image and then iteratively narrowing down to the correct location of the identified brain tumor. In order to achieve this localization, transformations are applied to the initialized bounding box that initially covers the whole image and is finally reduced to a tight bounding box. This sequence of transformations is decided by the agent that analyzes the content of the currently visible region to select the next best action. In theory each transformation should keep the object inside the visible region while removing as much background as possible. **Figure 1** illustrates some steps of this dynamic decision process. It is important to note, that this strategy does not follow a fixed path to each target. To stimulate the attention of the proposed agent, we use a reward function proportional to how well the current box covers the target lesion. A reward function is incorporated to learn the localization policy passed on the Deep Q-learning network algorithm (DQN) [20]. As a result, the trained agent can possibly localize a single instance of brain tumor, which means that the algorithm can correctly find a target after processing only a few regions of the image. We performed an experimental evaluation in the BraTS 2017-2020 dataset containing HGG and LGG brain tumors. We will now present the components of this proposed approach in the context of a RL problem formulation.

2 Preliminaries

2.1 Problem Formulation:

RL problems are formulated as an MDP, where $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \gamma)$. Here, states $s \in R^n$, actions $a \in \mathcal{A}$, state transition distribution $\mathcal{P} = \mathcal{P}\ (s_{t+1}, r_t \mid s_t, a_t)$, and discount factor $\gamma \in [0, 1)$. $\mathcal{S}$, $\mathcal{A}$, and $\mathcal{P}$ collectively define mechanics and rewards for a given task. Without prior knowledge of $\mathcal{P}$, RL algorithms use their experience to maximize expected rewards $\mathcal{R}$. In this case, the RL agent is learning from images and their pixels, and thus receives image-based observations $o_t = O_t(s_t) \in R^k$. Our proposed formulation considers a single image as the observation in the environment, in which the agent transforms a bounding box using a set of actions. The goal of the agent is to land a tight box in a target object lesion can be observed in the environment.

2.2 Localization Actions

The set of actions, $\mathcal{A}$, is a set of eight transformations which can be applied to the bounding box and one action to terminate the search process. Possible actions can be seen in Figure 2, and they include actions that

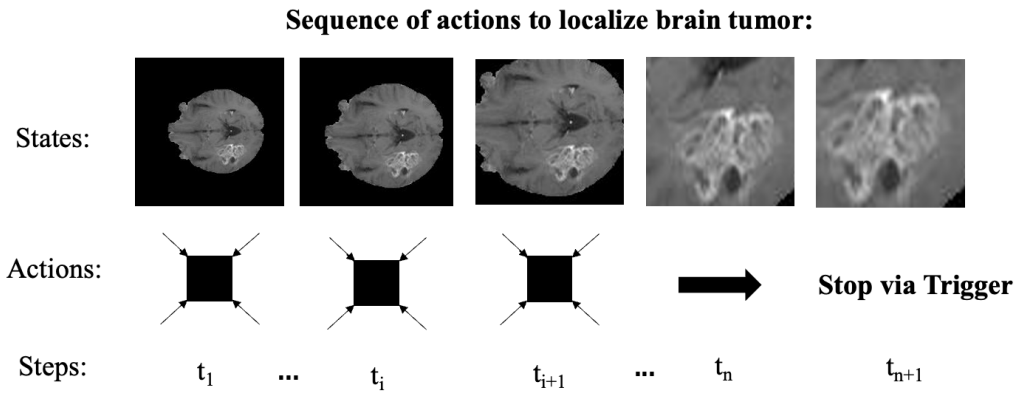

Figure 1: An example sequence of actions taken by the proposed method to localize a T_1-ce brain tumor. The RL algorithm bounds regions within the image and then decides how to transform the bounding box to progressively localize the tumor.

move the bounding box in the horizontal and vertical axes, actions to change scale, and actions to modify aspect ratio. Therefore, the agent has four degrees of freedom to transform the bounding box during an interaction within the environment. The bounding box is represented by the coordinates in pixels of its two corners of region— denoted: b = $[x_1, y_1, x_2, y_2]$. Note that the origin in the image plane is located in the top-left corner. Transformation actions make a discrete change to the box by a factor relative to its current size via (**Equation 1**). The new transformations are then obtained by adding or removing w or h to the x or y coordinates, depending on the desired effect. For example, to obtain a horizontal move to the right adds w to x_1 and x_y,. In this implantation α = 0.2 as it was reported to give a good trade-off between speed and localization accuracy [19]. It also was reported in exploration experiments that smaller values make the agent slower to localize objects, while larger values make it harder to place the bounding box correctly [19]. The only action that does not transform the bounding box is the trigger which indicates that an object is correctly localized over by the potential tumor by the current box. This action terminates the sequence of the current search. The trigger also modifies the environment by marking the region covered by the box with a black cross as shown in the final frame. This mark serves as an inhibition-of-return (IoR) mechanism where the currently attended region is prevented from being attended again.

$$\alpha_w = \alpha * (x_2 - x_1) \quad \alpha_h = \alpha * (y_2 - y_1) \tag{1}$$

2.3 Localization State:

The state representation is given by *o* and *h*, where *o* a vector of the observed region, and *h* is a vector with the history of taken actions. The set of possible states, $\mathcal{S}$, includes arbitrary boxes from a large set of images, and includes all combinations of actions that lead to those boxes. Therefore, it is essential to design a state representation. *O* is extracted from the current region using a pre-trained CNN seen in **Figure 3** [21]. We also expand the region to include 16 pixels of context around the original bounding box. The history vector, *h*, is a binary vector that informs which actions have been previously used in the past. The history vector encodes 10 past actions, which means that $h \in R^{90}$, where each action in *h* is represented by a nine-dimensional binary vector, where all values are zero except for the one corresponding to the action taken.

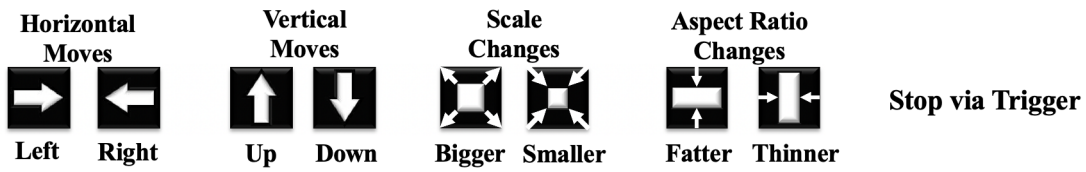

Figure 2: The nine possible actions the proposed MDP can take. This yields four degrees of freedom to the RL agent for the respective bounding.

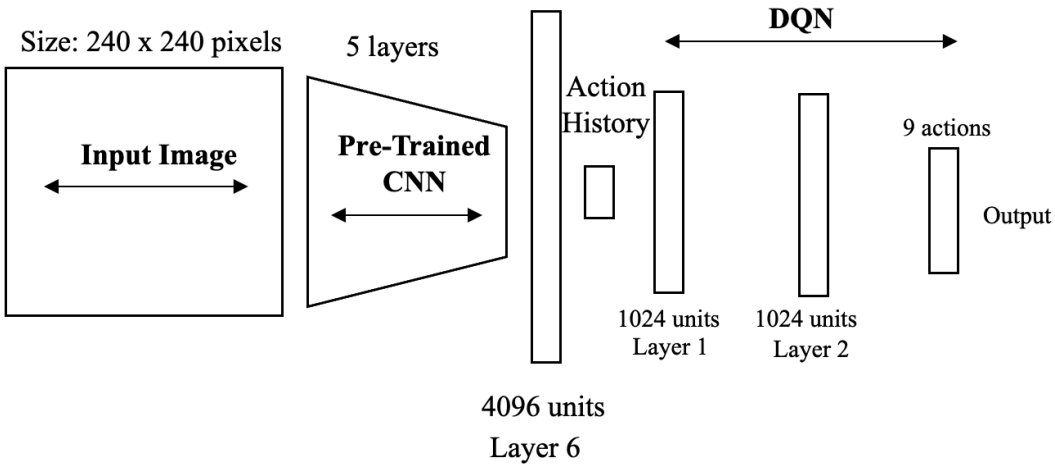

Figure 3: The Architecture of the DQN. It features the input image and a region which is processed by a pre-trained CNN with 5 convolutional layers and 1 fully connected layer. The output of the CNN is concatenated with the action history vector to complete the state representation. Finally, the state representation is processed by the Q-network which predicts the value of the nine actions.

2.4 Localization Reward:

The reward function, $\mathcal{R}$, can be considered proportional to the improvement that the agent makes to localize the brain tumor after selecting a particular action. Intersection-over-Union (IoU) between the target lesion and the predicted bounding box at any given time measures improvement at any given time. The reward function is estimated via the differential of IoU from one state to another (**Equation 2**). The reward function can be estimated during the training phase only because it requires ground truth boxes, g, to be calculated.

$$IoU(b, g) = area(b \cap g)/area(b \cup g) \tag{2}$$

$$R_a(s, s') = sign(IoU(b, g) \ - \ IoU(b, g)) \tag{3}$$

Equation 3 states that the reward is positive if IoU improved from state s to state s' and negative otherwise. This reward scheme is binary $r \in \{-1, +1\}$, and applies to any action that transforms the box. The agent pays a penalty for taking the box away from the target and is rewarded to keep the target object in the visible

region until there is no other transformation that improves localization. In that case, the best action to choose should be the trigger. The trigger has a different reward scheme because it leads to a terminal state that does not change the box, and thus, the differential of IoU will always be zero for this action. The reward for the trigger is a thresholding function of IoU (Equation 4). Here, ω is the trigger action, η is the trigger reward, and τ is a threshold that indicates the minimum IoU allowed to consider the attended region as a positive detection. The standard threshold for object detection evaluation mathτ = 0.5 during training to encourage better localization [19]. A larger value for beginmathτ has a negative effect in performance because the agent learns that only clearly visible objects are worth the trigger and tends to avoid truncated or naturally occluded objects. In our experiments η = 3.0. This proposed reward makes the agent follow a greedy strategy, where the agent prefers short sequences because any unnecessary step pays a penalty that reduces the accumulated utility.

$$R_w(s, s') = \begin{cases} +\eta & if\ IoU(b, g)\ \geq \tau \\ +\eta & otherwise \end{cases} \quad (4)$$

3 Methods

3.1 Deep Q-Learning for Object Localization:

In this implementation the overall goal of the agent is to transform a bounding box by selecting actions in a way that maximizes the sum of the rewards received during an interaction with the environment—this is referred to as an episode. The challenge is to define a policy that guides the decision-making process of this agent. A policy is a function π(s) that specifies the action a to be chosen when the current state is s. Since we do not have the state transition probabilities and the reward function is data-dependent, the problem is formulated as a reinforcement learning problem using Q-learning [22]. Here, we follow a DQN algorithm approach which estimates the action-value function using a neural network. After learning the action-value function Q(s, a), the policy that the agent follows is to select the action a with the maximum estimated value [20].

As discussed above, the DQN takes the state representation (Section 2.3) which gives an output in the form of the value attributed to one of nine actions (Section 2.4). The DQNs are trained with the pre-trained CNN architecture illustrated in Figure 3 [19]. The parameters of the DQN are initialized at random. Then, the agent is interacts with the environment in multiple episodes, each presenting a different training image. The policy followed during training is -greedy, which gradually shifts from exploration to exploitation according to the value of [22]. Normally during exploration the agent selection actions randomly to observe different transitions and collects a varied set of experiences, but here exploration does not proceed with random actions. Since the environment knows the ground truth boxes, and the reward function is calculated with respect to the IoU with the current box, we can identify which actions will give positive and negative rewards [23]. Then the agent to choose one random action from the set of positive actions. Since there might be multiple positive actions because there is no single path to localize an object the algorithm terminates in 15 epochs. The parameters are updated using stochastic gradient descent and the back-propagation algorithm, and we also use dropout regularization [24].

3.2 Training and Evaluation the Localization Agent:

Once an agent is trained with the procedure described above, it learns to attend regions that contain objects of the target brain tumor subtype. Here, we categorized brain tumors by three categories or subtypes from the BraTS 2017-2020 datasets—both HGG and LGG combined, only HGG, and only LGG. Since we do not know the number of tumors present in a single image beforehand, we let the agent run for a maximum of 50

steps, so only 50 regions are evaluated per test image. At each step, the agent decides to transform the current bounding box or selects the trigger to indicate that an object has been found. When the trigger is initialized the search for other objects continues from a new box that covers a large portion of the image. Training and Evaluation sets were performed with an 75% to 25% split relative to the amount of patient subject found within each category.

3.3 BraTS 2017-2020 datasets:

The images we used for training and testing came from the BraTS 2017-2020 HGG and LGG databases [25]. 888 and 300 HGG and LGG T_1-ce 3D brain volumes were selected from these datasets in total. Because we implemented a 2D localization model, each patient's 3D T_1-ce MRI was preprocessed to obtain a 2D axial slice. The central slice of each tumor was extracted, and MR signal intensity was normalized to be between [0,1].

Note: In the context of gliomas, Grades I and II are considered LGG, while grades III and IV are highly malignant and called HGG. LGG have been harder to detect in automated AI systems. LGG type I is generally benign and tends to remain unobservable and untreated in time [25].

4 Results

We focused on T_1-ce images because they represent the clinical gold-standard for glioma diagnosis. When applying our model to both the training and test data sets, it should be noted that the agent would run for the maximum of 50 allowed steps rather than converging on a solution by using the trigger action. We noticed that the agent most easily identified lesions when provided both HGG and LGG images as described in Table 1. This either indicates that using both channels provided a larger latent space for the model to navigate or that more data lends to improved performance. If the latter is the case we may not in reality reek the benefits of employing an RL model framework in this context, because for large datasets SDL and other machine-learning approaches may be more accurate and more efficient. We would follow up on these results by testing the models with comparable training data.

As shown in Figure 4, the agent is able to adequately localize the lesion within the image. In future experiments we would add padding around the bounding box to make a more generous threshold for correctly labeled tumors. The learning curves in Figure 5 show an increasing evaluation accuracy and max Q-value with a decreasing as a function of epochs. In future experiments, we would tune parameters and the reward function to augment the learning capabilities of the model.

Dataset	**Type**	**Correct**	**Incorrect**	**Total**	**Accuracy**
T_1-ce	**HGG_LGG**	130	167	297	30.4
T_1-ce	**HGG**	3	219	222	1.33
T_1-ce	**LGG**	2	73	75	2.60

Table 1: Deep Q-Network Performance.

5 Discussion

Although DRL is widely used in a wide variety of applications, such as robotics and computer games to our knowledge it has not been widely implemented in the field of radiology. This work represents a preliminary application of DRL within the context of identifying lesions from MRI brain images. We have applied DQN

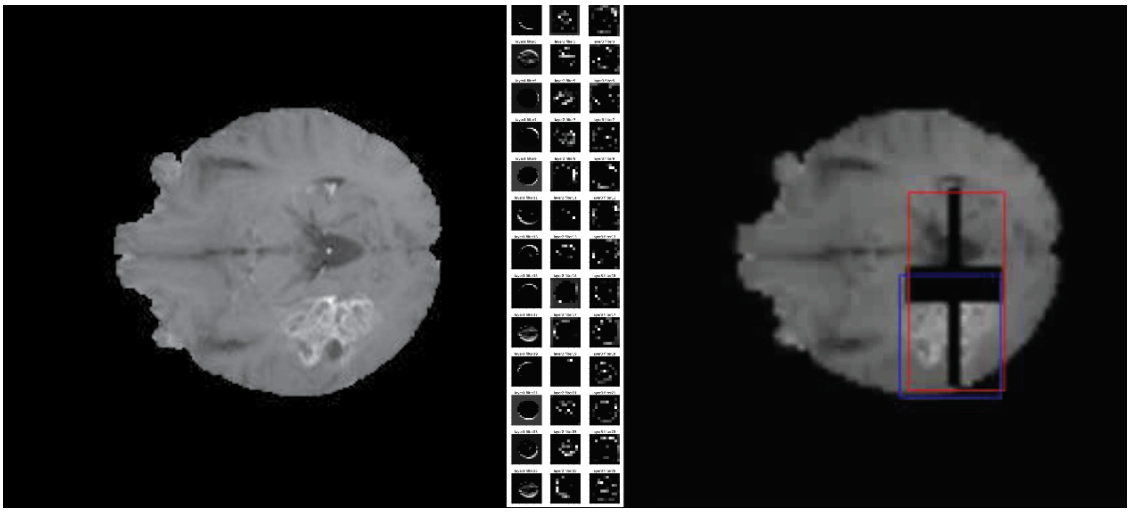

Figure 4: Agent attempting to localize brain lesion on T_1-ce when only the HGG dataset is provided. The left panel shows the original brain image. The middle panel shows a subset of the filters within the model. The right panel shows the agent's performance localizing the lesion.

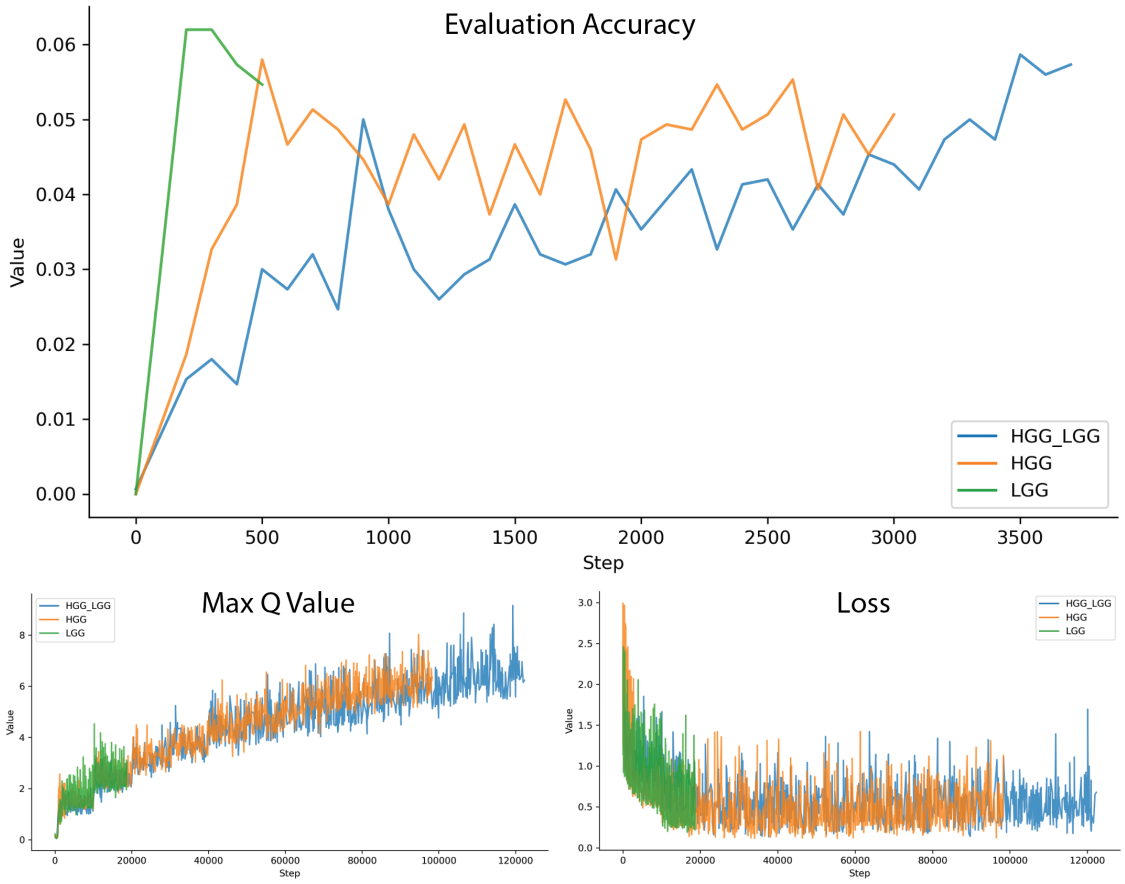

Figure 5: Deep Q-Network Learning Curves. The top panel shows evaluation accuracy during training. The lower two panels show the maximum Q value and loss for the model.

model to localize HGG and LGG brain tumors from the publicly available BraTS 2017-2020 image databases. Here, we were able to present a system that attempted to learn to localize tumors within (T_1-ce) 2D axial MRI images using an attention-action strategy. This approach demonstrated some efficacy in localizing the data when both HGG and LGG tumors were included in the training and test set, with an accuracy of 30.4% tumors correctly localized. And, little to no success when localizing HGG and LGG tumors alone in the respective training and test sets. We would like to acknowledge that this localization policy represents a challenging task since the tumors can be localized in different paths; and, the amount of training data used as inputs to this RL model is much smaller than recent SL and SDL approaches. For example, we did not employ data augmentation, a standard way to increase training set size used in SDL. In theory, if a similar model could be improved and optimized on relatively small datasets, RL approaches in medical imaging could be utilized at single institutions or for accelerating clinically-relevant research. Additionally, the notion that RL performs learning on all parts of an image, including the non-lesion portions, the algorithm should gain a more extensive sense of the image relative to SDL and other machine learning approaches. Thus, these approaches would be less sensitive to the random noise that is introduced when encountering images from a new scanner.

Nevertheless, our proposed agent did not learn and adequately optimize the policy to find lesions to a clinical standard. In future studies, we would first take greater care in optimizing the parameters mentioned above in the state and reward functions. We noticed that often the agent would continue to explore the image instead of converging to a target and effectively preform the trigger action. In this case we would aim to optimize τ as well as other parameters to increase the average accuracy of localization. In future studies we would also like to approach different MRI contrasts and see how image quality affects model performance. Finally, at some point this method would need to be extended to 3D MR brain volumes.

6 Member Contributions

Tanner and Lucas contributed equally in this project. Tanner acquired and preprocessed the BraTS datasets, as well as helped implement the proposed proposed RL model. Lucas also helped implement the proposed model and also performed analysis on the trained model. Over the course of the semester, we ran an equal amount of experiments, committed an equal amount of code, and all equally contributed to the final writing of the paper. The code used in this study is available through Google Drive.

7 References

1. Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A.A., Veness, J., Belle- mare, M.G., Graves, A., Riedmiller, M., Fidjeland, A.K., Ostro- vski, G., et al., 2015. Human-level control through deep reinforcement learning. Nature 518, 529.
2. Luketina, J., Nardelli, N., Farquhar, G., Foerster, J., Andreas, J., Grefenstette, E., Whiteson, S., Rockta schel, T., 2019. A survey of reinforcement learning informed by natural language. arXiv preprint arXiv:1906.03926.
3. Finn, C., Tan, X.Y., Duan, Y., Darrell, T., Levine, S., Abbeel, P., 2016b. Deep spatial autoencoders for visuomotor learning, in: Kragic, D., Bicchi, A., Luca, A.D. (Eds.), IEEE International Conference on Robotics and Automation, ICRA, pp. 512–519.
4. Bernstein, A., Burnaev, E., 2018. Reinforcement learning in computer vision, in: Tenth International Conference on Machine Vision (ICMV 2017), International Society for Optics and Photon- ics. p. 106961S.
5. Morgan P McBee et al. “Deep learning in radiology”. In: Academic radiology 25.11 (2018).
6. Luca Saba et al. “The present and future of deep learning in radiology”. In: European journal of radiology 114 (2019), pp. 14–24.

7. Spyridon Bakas et al. “Identifying the best machine learning algorithms for brain tumor segmentation, progression assessment, and overall survival prediction in the BRATS challenge”. In: arXiv preprint arXiv:1811.02629 (2018)
8. Thomas Weikert et al. “A Practical Guide to Artificial Intelligence–Based Image Analysis in Radiology”. In: Investigative radiology 55.1 (2020), pp. 1– 7.
9. Hosny A, Parmar C, Quackenbush J, Schwartz LH, Aerts HJWL. Artificial intelligence in radiology. Nat Rev Cancer. NIH Public Access; 2018;18(8):500–510.
10. Choy G, Khalilzadeh O, Michalski M, et al. Current applications and future impact of machine learning in radiology. Radiology. Radiological Society of North America; 2018;288(2):318–328.
11. Stember, J.N., Shalu, H. (2020). Deep reinforcement learning to detect brain lesions on MRI: a proof-of-concept application of reinforcement learning to medical images. ArXiv, abs:2008.02708.
12. Goodfellow IJ, Shlens J, Szegedy C. Explaining and harnessing adversarial examples. arXiv Prepr arXiv14126572. 2014.
13. Holzinger A, Langs G, Denk H, Zatloukal K, Müller H. Causability and explainability of artificial intelligence in medicine. Wiley Interdiscip Rev Data Min Knowl Discov. Wiley Online Library; 2019;9(4):e1312.
14. Stember, J.N., Shalu, H. (2021). Deep reinforcement learning-based image classification achieves perfect testing set accuracy for MRI brain tumors with a training set of only 30 images. ArXiv, abs:2102.02895.
15. Ali I, Hart GR, Gunabushanam G, Liang Y, Muhammad W, Nartowt B, Kane M, Ma X and Deng J (2018) Lung Nodule Detection via Deep Reinforcement Learning. Front. Oncol. 8:108. doi: 10.3389/fonc.2018.00108.
16. P.Sermanet,D.Eigen,X.Zhang,M.Mathieu,R.Fergus,and Y. LeCun. Overfeat: Integrated recognition, localization and detection using convolutional networks. arXiv:1312.6229, 2013.
17. K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. arXiv:1409.1556, 2014.
18. D. Erhan, C. Szegedy, A. Toshev, and D. Anguelov. Scalable object detection using deep neural networks. In CVPR. IEEE, 2014.
19. Caicedo, J.C., Lazebnik, S. (2015). Active Object Localization with Deep Reinforcement Learning. 2015 IEEE International Conference on Computer Vision (ICCV), 2488-2496.
20. V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, et al. Human-level control through deep rein- forcement learning. Nature, 518:529–533, 2015.
21. M. D. Zeiler and R. Fergus. Visualizing and understanding convolutional networks. In ECCV. Springer, 2014
22. R. S. Sutton and A. G. Barto. Reinforcement learning: An introduction. MIT press, 1998.
23. S. Levine, C. Finn, T. Darrell, and P. Abbeel. End-to-end training of deep visuomotor policies. arXiv:1504.00702, 2015.
24. N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: A simple way to prevent neural networks from overfitting. The Journal of Machine Learning Research, 15(1):1929–1958, 2014.
25. Menze BH, Jakab A, Bauer S, et al. The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS). IEEE Trans Med Imaging. NIH Public Access; 2015;34(10):1993-2024.